

Seeking my first professional opportunity, I am eager to apply my skills in a stimulating industrial environment. My professional goal is to combine technical expertise with creativity in order to bring a personal touch and added value to the projects I contribute to.

Formation

- 2021-2026 **Industrial Engineering (spec. in electromechanics) - Distinction**
ECAM Brussels Engineering School, Woluwe-Saint-Lambert
- 2021-2025 **Art education : Computer graphics and digital arts**
Académie des beaux-arts, Watermael-Boisfort
- 2015-2021 **CESS (Science-History options)**
Institut Saint-André, Ixelles

Languages

French, English (B2)

Experiences

- 01-06/2026 **Thesis – Design and implementation of interactive fountains integrated into the “Mecalodon” attraction layout.**
Walibi Belgium, Wavre
- Full development of an interactive animation: analysis of existing technologies and concept design
 - Preparation of electrical schematics, short-circuit calculations, cable sizing, and protection selection
 - Design of the electrical control cabinet and selection of components
 - Selection of pumps, pressure loss calculations, and hydraulic network design
 - Development of control and synchronization sequences for aquatic effects (DMX, 24 V, etc.)
 - Preparation of technical drawings and layout plans for on-site installation
 - Coordination with suppliers and technical teams: site surveys, assembly, wiring, and installation
- 03-05/2026 **Design and implementation of a manufacturing and inspection procedure for mechanical parts**
Walibi Belgium, Wavre
- Analysis of critical roller coaster components (INTAMIN, MACK Rides, Gerstlauer) based on technical drawings and maintenance data
 - Definition of technical and regulatory requirements, including EN 13814
 - Consideration of manufacturing constraints, inspection requirements, and material selection
 - Design of mechanical part drawings using SolidWorks, including dimensional and geometric tolerances, fits, and manufacturing constraints
 - Collaboration with laboratories and subcontractors for chemical analyses, metallographic examinations, hardness measurements
 - Material characterization, heat treatment analysis, and assessment of associated mechanical properties
- 2023-2026 **Section Representative – Electromechanics**
ECAM Brussels Engineering School, Woluwe-Saint-Lambert
- Conducted regular surveys and gathered students' questions, feedback, and concerns
 - Identified areas for improvement and communicated with the teaching staff to help address students needs
- 2023-2025 **Private lessons**
MySherpa, Bruxelles
- Providing support to high school students in mathematics, physics, and study skills

Personal projects

- Since 2022 **Okbots Expedition**
Development of a video game using Unreal Engine and various production software. The graphic design, programming, gameplay, and sound design were independently developed. The project incorporates various sequential logic systems inspired by industrial automation, including drone-based object tracking, the management of physical and dynamic properties, as well as a rotation system based on real-time calculation of ground orientation.
- Since 2020 **Créations graphiques et illustrations**
Production of original illustrations with a unique graphic style and development of visuals adapted to different creative projects.
More informations: aurelienpetit.com

Professional software

Technical aspect : Solidworks (CSWA), Matlab, Abaqus, TIA Portal, RobotStudio, Eleccalc, Fusion, etc.
Creative aspect : Suite Adobe, Affinity, Final Cut Pro, Motion, Logic Pro, Unreal Engine 5, etc.

Interests

Technological innovation, theme parks, drawing, space exploration, tennis, orchestral music, creative and management-focused video games, economics, and scale modelling

